

The Northern Sea Route traffic visualization

Goal

To create a visual product that shows the amount of traffic via Russia's Arctic seas in a clear way for a wide audience.

Outputs

1. **An interactive map** showing:

- The amount of traffic in the last available year
- A change in the amount of traffic in the last year compared to previous years
- Protected natural areas in Russia's Arctic seas and along the Arctic coastline

Interactive elements:

- As one idea, the map shows traffic by default for the last available year. If a user moves a slider, they can see the amount for previous years.
 - A user can filter traffic by type of vessel, age, type of fuel burnt, ALL, etc.
- ### 2. **An internal spreadsheet** with data: number of **vessels/voyages** each year. This raw data will allow us to make conclusions/infographics in our report, such as: *"In 2024, the number of voyages via Russia's Arctic increased by XX %", etc.*

It would be useful to have this data in dataset (if they are available):

- Name and type of each vessel (oil tanker, nuclear icebreaker, LNG carrier, etc.)
- Number of vessels that travelled via NSR each year
- Distance traveled by each vessel
- Route of each vessel
- Age of each vessel

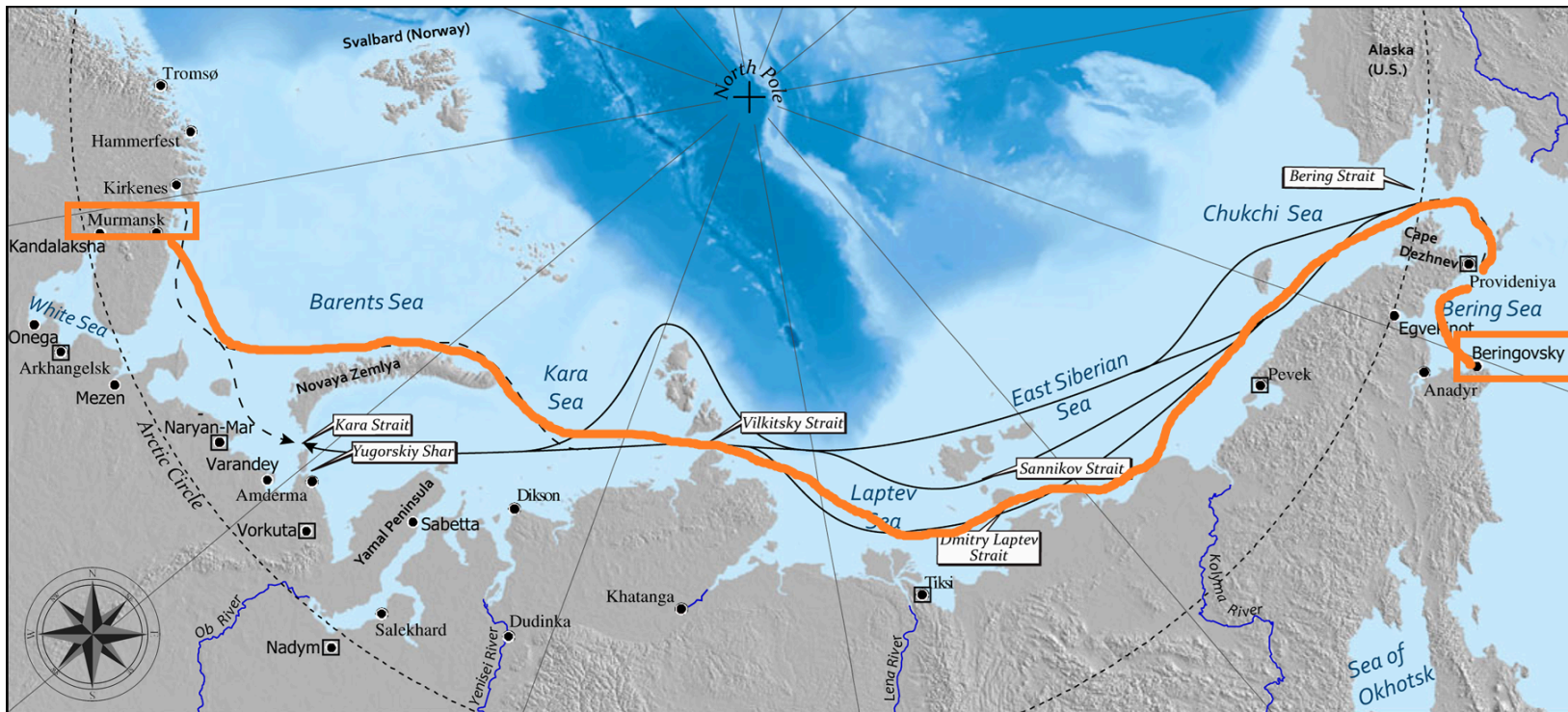
- Fuel used by each vessel: does it burn heavy fuel oil (HFO), other oil types, LNG, etc.

Focus

Geographically

The entire Russian coastline along the Arctic Ocean (from Murmansk port to Beringovskiy port):

- the "official" Northern Sea Route's seas (the Kara Sea, Laptev Sea, East Siberian Sea, and Chukchi Sea)
- + the Barents Sea
- + the Bering Strait, the Bering Sea, Provideniya port, Beringovskiy port



Time

The period for which data is available.

As one idea, the interactive map could show traffic by default for the last available year. If a user moves a slider, they can see the amount for previous years.

Sources

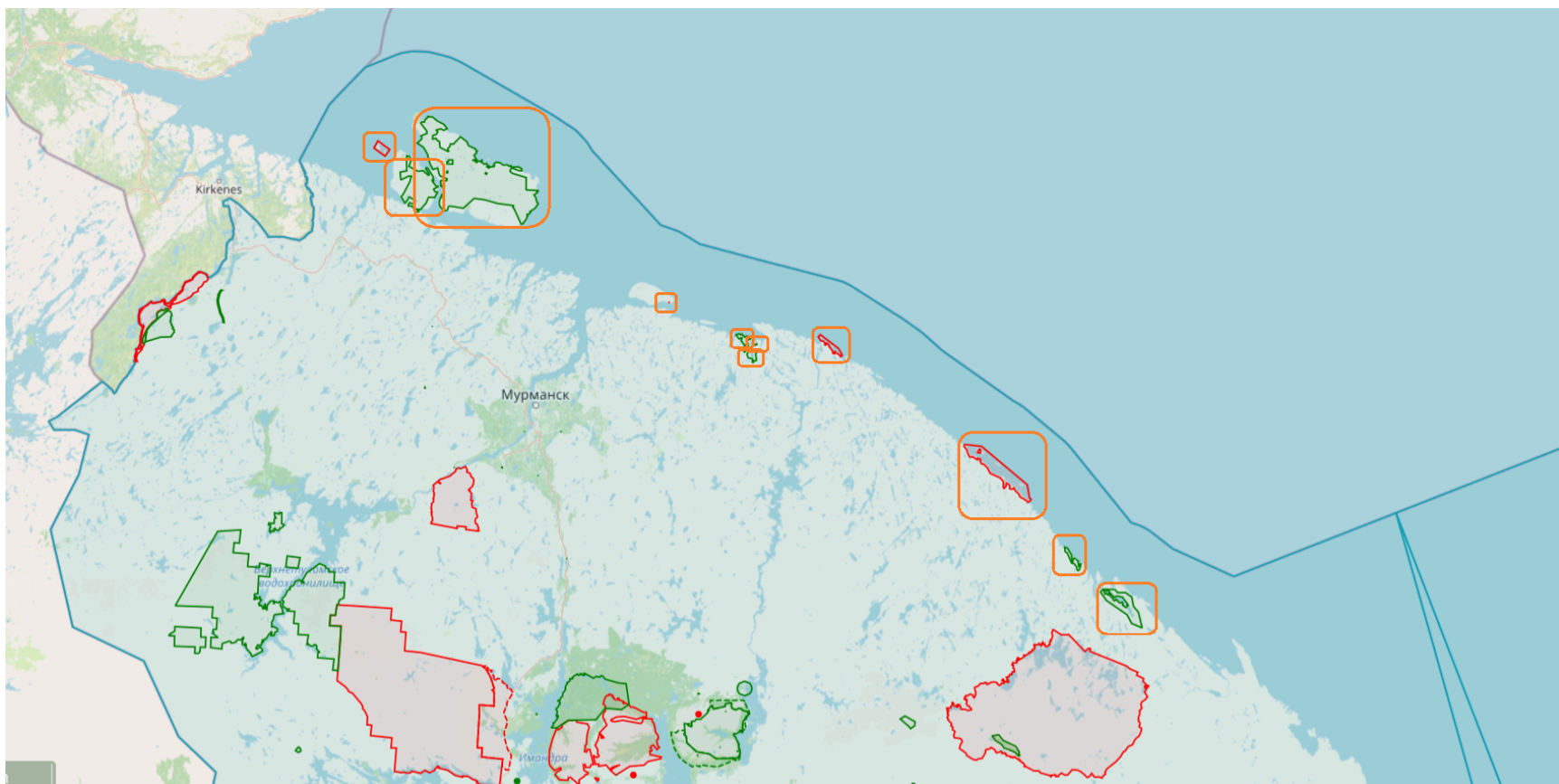
1. <https://globalfishingwatch.org/>
2. <https://pame.is/ourwork/arctic-shipping/current-shipping-projects/astd/> (paid)
3. **Protected areas in the Arctic:** <https://xn--80aa2azak.xn--g1agk6a.xn--p1ai/>.
 - After clicking on a protected area, shape files are displayed in the right-bottom corner with the button *Скачать* (*Download*, see screenshot below). Protected areas in red have a legal status of federal level, and green ones — of a regional level (we are interested in both of them). For the purpose of this product, we need only areas located either in the Arctic Ocean (like archipelagos) or along Russia's Arctic coastline, so that border crosses the sea.



ООПТ	Орган управления ООПТ	Статус ООПТ	Дата добавления	Дата обновления	SHP
о	Федеральное государственное бюджетное учреждение "Объединенная дирекция заповедников Таймыра"	действующий	2024-10-10	2024-10-24	Скачать

- For this product, we need only areas located either **in the Arctic Ocean** (like archipelagos) or **along the Arctic coastline** (so their border crosses the sea). For example, if we zoom in on the western corner of the map, we need only the protected areas circled below in orange.

Note: some can be quite small, so a greater zooming is needed in order not to miss them. :)



- Every protected area has a date of its establishment (box *Дата учреждения*, see screenshot below). If the shipping data on our map spans a number of years and the user can go back in time, it will be good to make natural areas appear on the screen only in those years when the specific natural area was already established. For example, if it was established, say, in 2022, it won't be shown in years before 2022.

ООПТ

Числовой идентификатор ООПТ	Название	Значение ООПТ	Категория ООПТ	Дата учреждения
81	Русская Арктика	Федеральный	национальный парк	2009-06-15

Reference

A static map which shows changes in traffic in an easy-to-grasp way:

Figure 1: Shipping Traffic of Carrying and Using HFO from 2021 to 2023

